Renewal Process for Wildfire Spreading

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The occurance of wildfire, which is modeled with the probability, can be viewed as a kind of point processes, especially the renewal process. The renewal process refers to the point process whose inter-point time interval distribution is not i.i.d.(independent and identically distributed).

1 Theory

1.1 Renewal Process

Consider two different incidents of a point process which is occured at different time points t_0 and t, where $t_0 < t$. Define the followings:

1. **next point distribution** $p(t|t_0)$**:** the probability distribution for time point of the next event, given event at t_0,
2. **survivor function** $S(t|t_0)$**:** the probability of not-occurance of event until time t, given event at t_0, and
3. **hazard function** $\rho(t|t_0)$**:** the probability distribution for the chance of occurance of event at time point t, given event at t_0.

Observe that the following equations are plausible:

$$p(t|t_0) = S(t|t_0)\rho(t|t_0), \qquad S(t|t_0) = 1 - \int_{t_0}^{t} p(\tau|t_0)\mathrm{d}\tau.$$

From these two equations, the differential equation

$$\frac{\partial}{\partial t} S(t|t_0) = -S(t|t_0)\rho(t|t_0)$$

can be established. Since $S(t_0|t_0) = 1$, solving the differential equation, we have

$$S(t|t_0) = \exp\left[-\int_{t_0}^{t} \rho(\tau|t_0)\mathrm{d}\tau\right],$$

and hence

$$p(t|t_0) = \rho(t|t_0)\exp\left[-\int_{t_0}^{t} \rho(\tau|t_0)\mathrm{d}\tau\right].$$

Note that if $\rho(t|t_0)$ does not depend on t_0, that renewal process is Poisson process.